

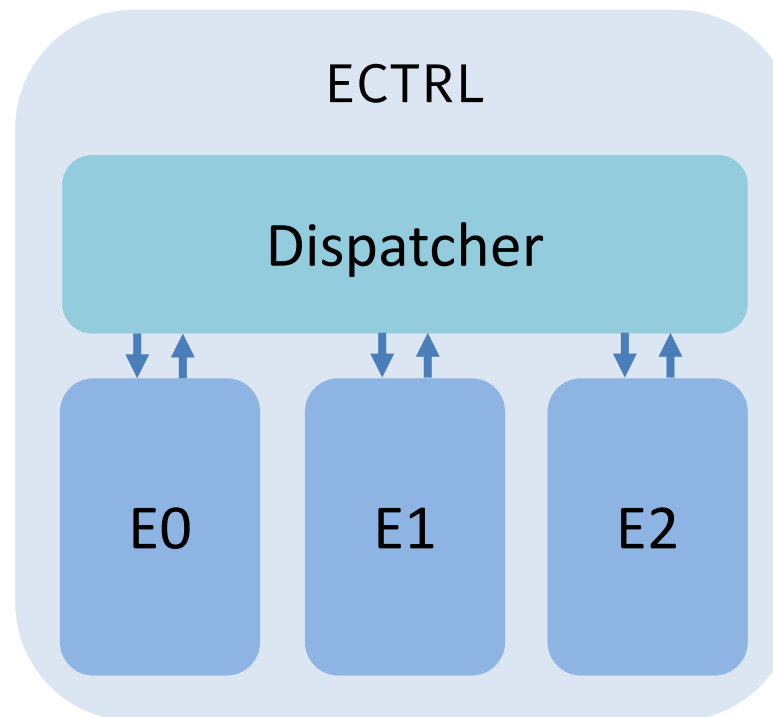


HW03 Code Review

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Modularity Design

- Each elevator shares the same implementation
- Dispatcher manages hall requests & allocates to elevators



Modularity Design

- Write sub-module for an elevator
- Instantiate sub-module inside main module

```
ELEVATOR_CORE E0 (  
  .clk(clk),  
  .rst_n(rst_n),  
  .car_req(car_req[7:0]),  
  .up_req(up_req_0),  
  .down_req(down_req_0),  
  .btn_open(btn_open[0]),  
  .btn_close(btn_close[0]),  
  .is_full(is_full[0]),  
  .cur_floor(cur_floor[2:0]),  
  .lift_status(lift_status[2:0]),  
  .serve_dir(serve_dir[1:0])  
)
```

Request signals from dispatcher

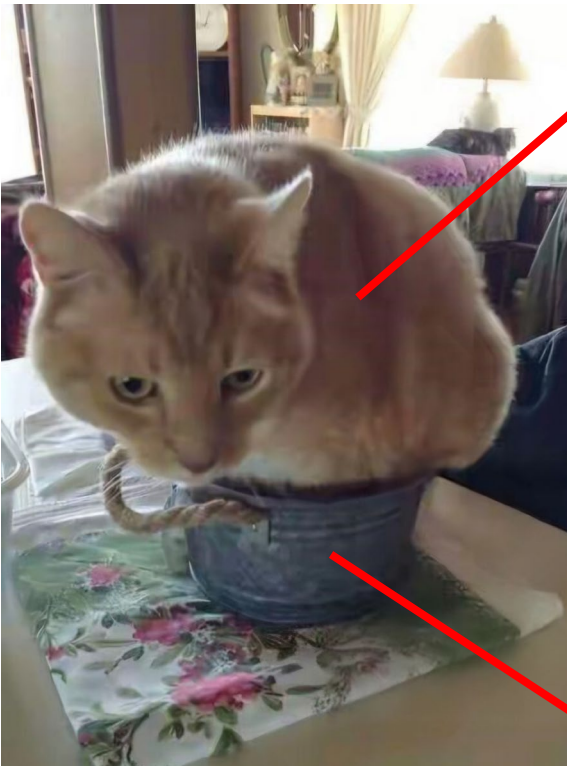
```
module ELEVATOR_CORE (  
  input clk,  
  input rst_n,  
  
  input [7:0] car_req,  
  input [7:0] up_req,  
  input [7:0] down_req,  
  input btn_open,  
  input btn_close,  
  input is_full,  
  
  output logic [2:0] cur_floor,  
  output logic [2:0] lift_status,  
  output logic [1:0] serve_dir  
);  
  
...  
  
endmodule
```

Algorithm

- First-Come, First-Served (FCFS)
 - Processes requests in strict chronological order
- SCAN (Elevator Algorithm)
 - Travels continuously in one direction & reverses only at the top or bottom floor
 - Serves all requests along the path
- LOOK Algorithm
 - "Looks ahead" for pending requests
 - Reverses immediately if the path ahead is clear
- Zoning / Sectoring
 - Divides the building into specific service zones
 - Assigns elevators to dedicated floors
- Etc.

Timing Slack

- DO NOT wire input to output using pure combinational logic
 - input external delay = output external delay = half cycle time



combinational logic
from input to output

```
assign is_full = {&(pax_cnt[11:8]), &(pax_cnt[7:4]), &(pax_cnt[3:0])};
```

```
always_ff @(posedge clk or negedge rst_n) begin
  if (~rst_n) begin
    is_full <= 3'b000;
  end else begin
    is_full <= is_full_next;
  end
end

assign is_full_next = {&(pax_cnt[11:8]), &(pax_cnt[7:4]), &(pax_cnt[3:0])};
```

available time = 0 ns (= cycle time – input/output external delay)